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CSE 350: Digital Electronics and Pulse techniques

Exp-05: Implementing NMOS logic gates

Name:	Section:
ID:	Group:

Objectives

1. Construct NMOS logic gates.
2. Understanding the NMOS logic circuit operations.

Equipment and component list

Equipment

1. Digital Multimeter
2. DC power supply

Component

- NMOS IC IRF540 - x2 pieces
- Resistors -
    - 5 KΩ - x1 piece

Task 01: NMOS Inverter

THEORY

MOSFETs have gained popularity in logic circuits due to their high fabrication density and low power dissipation. When used in logic circuits, there are cases where only p-channel or only n-channel devices are employed. These are known as PMOS and NMOS logic circuits, respectively. An inverter circuit outputs a voltage representing the opposite logic-level to its input. Inverters can be constructed using a single NMOS transistor or a single PMOS transistor coupled with a resistive load or enhancement/depletion load.
The NMOS inverter circuit with resistive load is implemented as shown in Figure 01.

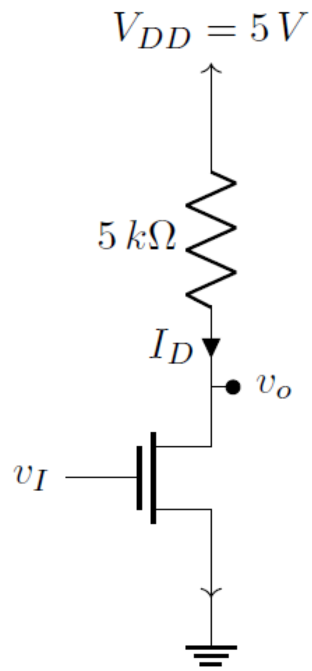


Figure 1 : NMOS Inverter with resistive load

Here, the input voltage V_I is applied to the gate terminal of the mosfet M1 and the output voltage V_O is taken from the drain terminal. The source terminal is directly connected to the ground. When the input V_I is LOW (0V), there is no channel formation from Drain to Source in the Enhancement type mosfet and thus it operates in the cutoff mode, so I_D=0 mA. As a result, there is no voltage drop across R_D and the output voltage V_O will be the same as V_{DD} = 5V (HIGH).
Conversely, if a HIGH input (5V) is applied as the input voltage V_I, the mosfet is assumed to be operating in the Triode mode. In this mode, I_D is governed by the equation,

$$I_D = K_n[2\,(v_{GS} - V_{TN})v_{DS} - v_{DS}^2]$$

Consequently, there will be a voltage drop across R_D and thus the output voltage V_O will be LOW and close to 0V. Thus, the output of the circuit is always the inverse of the input.

<u>Task 02: NMOS Inverter with Enhancement type load</u>

In this NMOS inverter, a MOSFET replaces the resistive load, greatly improving the packing density. The two MOSFET's are fabricated with identical thresholds and process transconductance parameters, for simplicity and high circuit yield. The circuit implementation is shown in Figure 02.

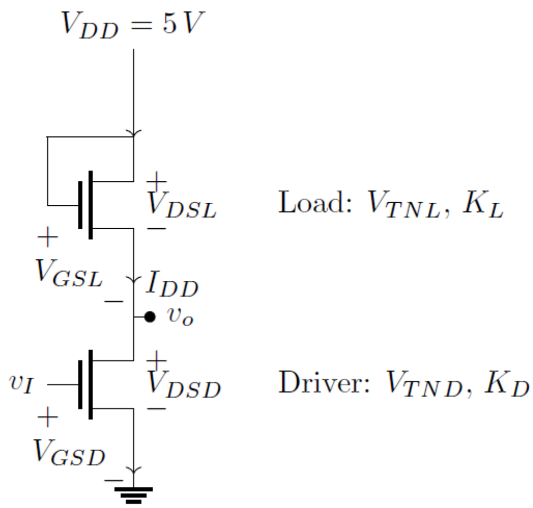


Figure 2 : NMOS inverter with Enhancement type load

The upper MOSFET is called the *load* whereas the lower MOSFET is the *driver*. We can see that, $V_{GL} = V_{DL} = V_{DD}$ for the *load* NMOS. Therefore,

$$V_{DSL} = V_{DL} - V_{SL} = V_{GL} - V_{SL}$$

$$Thus,\ V_{DL} - V_{SL} \geq (V_{GL} - V_{SL}) - V_{TNL}$$

So, the *load* NMOS is always in saturation regardless of the input and output voltage. When V_I is LOW (0V), no channel forms from the *driver* Drain to Source and the *driver* NMOS is in *cutoff* mode so I_{DD} = 0 mA. The *load* NMOS on the other hand is in saturation mode as proven earlier, but I_{DL} = 0 mA as well since the *driver* is in cutoff and all drain currents are zero. So we get $V_O = V_{DD} - V_{TNL}$. Conversely, when V_I is HIGH (5V), the channel forms from the *driver* Drain to Source and the *driver* NMOS is assumed to be in *triode* mode. The *load* NMOS on the other hand is in saturation mode as proven earlier. Using KCL, we get $I_{DL} = I_{DD}$.

$$K_{nL}(V_{GSL} - V_{TNL})^2 = K_{nD}[2(V_{GSD} - V_{TND})V_O - V_O^2]$$

Thus we get an output voltage V_O which will be LOW and close to 0V. Thus, the output of the circuit is always the inverse of the input.

Task 03: NMOS NOR gate with resistive load

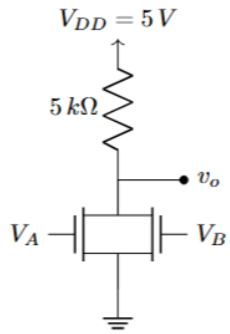


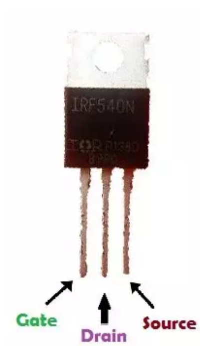
Figure 3 : NMOS NOR gate with resistive load

The given circuit is a **2-input NMOS NOR gate with a resistive load**. The resistor R is connected between the supply voltage V_{DD} (logic high) and the output node. The two NMOS transistors are connected in parallel between the output and ground, with their gates connected to inputs A and B, respectively. When both inputs A and B are at logic 0, both NMOS transistors are in cutoff mode, and the output is pulled high through the resistor R, resulting in a logic 1 at the output. If either A or B is at logic 1, the corresponding NMOS transistor turns ON, creating a conducting path to ground. This pulls the output low, resulting in a logic 0. If both A and B are 1, both transistors conduct, and the output remains low. Thus, the output is high only when both inputs are low, implementing the logic function of a NOR gate.

Procedure:

1. Connect the circuit as shown in Fig: 1, 2 & 3.
2. Observe the output for all possible input combinations and thus verify the type of gate.
3. Fill up the following tables. You must calculate the current values.

The pin diagram of a MOSFET is shown below:



Precautions:

1. When conducting the experiments please make sure to use the DC Power Supply and limit the current in the range of 0.1-0.3 A.
2. When measuring the current using the Ammeter, please make sure to place the red knob of the probe into the 10 A Max input of the multimeter.
3. The MOSFETS can get quite hot during the experiment so please take the values as soon as you can and **DO NOT TOUCH THE MOSFETS** with your bare hands and allow them some time to cool down.

Data Tables:

Table 1: NMOS Inverter with resistive load

$V_I(V)$	$V_{RD}(V)$	$I_{RD}(mA)$	$P_{RD}(mW)$	$P_{Total}(mW)$	$V_O\ (V)$

Table 2: NMOS Inverter with Enhancement type load

$V_I(V)$	$V_{GSL}(V)$	$V_{GSD}(V)$	$I_{DD}(mA)$	$P_{Total}(mW)$	$V_O\ (V)$

Table 3: NMOS NOR gate with resistive load

$V_A(V)$	$V_B(V)$	$V_R(V)$	$I_R(mA)$	$P_{Total}(mW)$	$V_O\ (V)$

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Signature

Report

Please answer the following questions briefly in the given space.

1. Using the value of I_D you calculated in table 1 for the NMOS inverter with resistive load, calculate the value of K_n for that NMOS. Given that $V_{TN} = 2\,V$

2. Design an NMOS NAND gate with resistive load.

3. For the NMOS inverter with resistive load and V_{DD} = 3V. Assume transistor parameters of K_n = 0.15 $\mu A/V^2$, and V_{TN} = 0.5 V. Find the value of R_D such that V_O = 0.1 V when V_I = 3 V.

4. Write a discussion and include the following: your overall experience, accuracy of the measured data, difficulties experienced, and your thoughts on those.